Koushik Sivarama Krishnan

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LinkedIn Github Portfolio

Highlights

- Master's in Applied Data Science with **3+ years** of experience using **Python**, **SQL**, and statistical modeling to clean complex data, build reliable features, and support production analytics and machine learning workflows.
- Partnered with engineers, researchers, and business stakeholders to investigate ambiguous data issues, define trustworthy metrics, and turn messy operational data into decision-ready dashboards, APIs, and reproducible analysis pipelines.
- Specialized in **data quality**, validation, and experimentation; built automated evaluation and monitoring frameworks that improved model reliability, reduced debugging time by **40%**, and increased trust in production decision support systems.

Skills

Programming	**Python**, **SQL**, Pandas, NumPy
Data Quality	data cleaning, data validation frameworks, missing value handling, outlier detection, inconsistency checks, feature engineering
Statistical Analysis & ML	statistical analysis, experimentation, hypothesis testing, A/B testing, causal inference, Logistic Regression, Random Forest, XGBoost, CatBoost, SHAP
Data Visualization	Power BI, Matplotlib, Seaborn, stakeholder-ready dashboards, KPI reporting
Data Engineering	PostgreSQL, ETL/ELT, PySpark, Dagster, data pipelines, event logging, telemetry
Collaboration & Documentation	cross-functional collaboration, metric definition, authoritative documentation, reproducible workflows, self-serve reporting

Experience

Co-op Data Scientist *09/2024 - 08/2025*

Insurance Corporation of British Columbia (ICBC) *Vancouver, BC*

- Built an Agentic AI system of claims knowledge retrieval by deploying an **AKS-hosted RAG assistant** built with **FastAPI and LangChain** that serves **500+ queries/day** for **100+ claims adjusters**, cutting manual search time by **60%**.
- Increased reliability of ML workflows by building an automated evaluation pipeline in Python with **PostgreSQL**-backed logging that tracked accuracy, faithfulness, and latency across **3** production releases, reducing model debugging time by **40%**.
- Delivered a **CatBoost-based risk-scoring API** (Flask + PostgreSQL) that prioritizes **1,200+ cases/month**, improving triage efficiency by **15%** through explainable decision support embedded into case management workflows.

Research Engine Developer *05/2024 - 02/2025*

Justice Data and Design Lab Victoria, BC

- Doubled downstream QA chatbot effectiveness by redesigning a **RAG-based legal question-answering pipeline** that improved relevant document retrieval accuracy by **2x**, integrating **LLM-based reranking** and cleaner text preprocessing for more reliable decision support.
- Operationalized data analytics automations by building end-to-end Python/SQL pipelines for **multiple MITACS-backed studies**, enabling projects to be re-run in **one command** with versioned data cleaning, transformations, and metrics.
- Surfaced high-impact process improvement opportunities by running large-scale text mining and topic modeling over **Reddit and legal education datasets**, delivering interactive **Power BI** dashboards that highlighted **top emerging issue clusters** and informed outreach and intake workflows.

Founding Machine Learning Engineer *04/2023 - 07/2023*

SeiSei.ai *India*

- Built and validated an end-to-end Hindi voice pipeline that improved FreeVC performance by **15%** through benchmarking, preprocessing, and data quality checks across noisy audio inputs.
- Automated collection, preprocessing, quality checks, and batch inference for **5,000+ hours** of audio data, creating cleaner training inputs and more reliable model iteration for deployment.
- Coordinated stakeholder feedback and deployment through Kubernetes on Truefoundry, translating ambiguous product requirements into monitored ML workflows with fail-safe evaluation practices.

Computer Vision Intern *09/2021 – 02/2022*

Drive Analytics *India*

- Built and deployed production video analytics pipelines on AWS using Celery and RabbitMQ, processing **1̃0K clips/day** and returning outputs within a **2-hour** delivery window for external partners.
- Investigated data quality failures across multi-view footage, low-resolution clips, occlusions, and weather noise, improving output trustworthiness by reducing misleading analytics in real production workflows.
- Evaluated and iterated on computer vision approaches against real-world failure cases, helping ensure analytics were reliable enough for downstream coaching and analyst use under strict turnaround requirements.

Deep Learning Intern *12/2020 – 06/2021*

MURF.ai India

- Delivered a **FastSpeech-2** voice cloning system by building an end-to-end audio data pipeline (preprocessing with **librosa**, feature engineering, training/inference workflow), enabling reproducible generation and faster iteration on voice quality.
- Reduced domain-specific speech recognition errors by fine-tuning STT models in **Azure Speech Studio**, improving WER through targeted data augmentation and acoustic feature optimization for noisy/real-world audio.

Education

Master of Engineering in Applied Data Science, *University of Victoria* **Victoria, BC** *2023-2025*

Notable Coursework: Optimization for Machine Learning, Scaling Large Language Models on GPUs, Data Privacy

Bachelor of Engineering in Computer Science and Engineering, *Anna University* **India** *2019-2023*

Notable Coursework: Artificial Intelligence, Cloud Computing, Data Structures, Data Warehousing, and Data Mining

Projects

Telecom Customer Churn & Upsell Prediction System *11/2025*

- Built an end-to-end churn prediction and customer upsell system on a **7,043-customer** telecom dataset, performing data preprocessing, feature engineering (**30+ features**), and statistical modeling using **Python, Scikit-learn, and XGBoost** to identify high-risk customers and cross-sell opportunities.
- Developed multiple predictive models (Logistic Regression, Random Forest, XGBoost) with comprehensive model evaluation using precision, recall, F1-score, and AUC-ROC metrics, achieving improved identification of high-risk churners in the top 10% segment through probability calibration and threshold optimization.
- Delivered an interactive **Streamlit dashboard** displaying portfolio-level churn risk by contract type and region, plus individual customer churn probability and upsell scores with **SHAP-based driver analysis**, providing actionable insights for retention and revenue optimization strategies.

NYC 311 Service Demand Forecasting & Optimization System *11/2025*

- Designed end-to-end machine learning system for **multi-dimensional demand forecasting** using NYC 311 service data, applying time-series analysis and predictive modeling to forecast service volumes across 10 categories, directly applicable to banking operations such as call center staffing and credit demand prediction.
- Implemented **Temporal Fusion Transformer** and Prophet models for hourly demand forecasting, achieving 23% lower MAPE than baseline through feature engineering (time-of-day, seasonality, service mix), model validation, and hyperparameter optimization, with attention mechanisms revealing key demand drivers for business interpretation.
- Developed **Bayesian optimization framework** for workforce allocation, reducing simulated wait times by 34% and labor costs by 18%; delivered interactive Streamlit dashboard for real-time scenario analysis and Pareto-optimal trade-off visualization between service quality and operational efficiency.

Publications

ORCID ID:0000-0003-0525-0677 | **Google Scholar**

- "Vision transformer based COVID-19 detection using chest X-rays"
- "SwiftSRGAN–Rethinking Super-Resolution for Efficient and Real-time Inference"
- "Benchmarking Conventional Vision Models on Neuromorphic Fall Detection and Action Recognition Dataset"
- "Efficient Super-Resolution For Chest X-rays"
- "MFAAN: Unveiling Audio Deepfakes with a Multi-Feature Authenticity Network"